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RAVENSBURG-WEINGARTEN
UNIVERSITY
OF APPLIED SCIENCES

Electronic Circuit Design

Lecture notes

Chapter 10: From Sensor to Digital Value

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Contents

10 From Sensor to Digital Value **3**
10.1 The Measurement Chain . . . 3
10.2 Stage 1: Signal Conditioning . . . 4
10.3 Stage 2: Anti-Aliasing . . . 6
10.4 Stage 3: Sample and Hold . . . 7
10.5 Stages 4 and 5: Flash ADC, Encoder and Display . . . 8
10.6 The Complete Chain in Falstad . . . 9
10.7 Exercises . . . 10

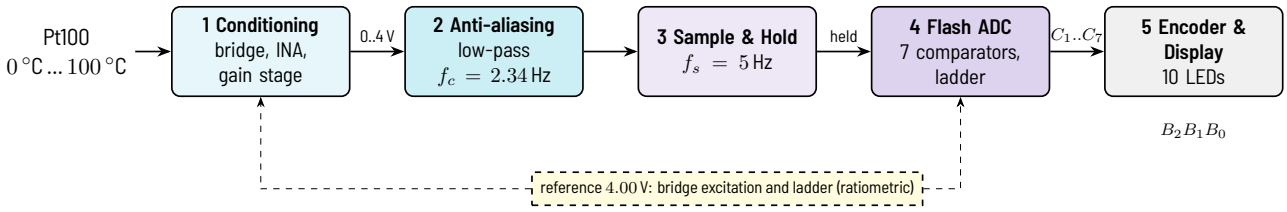
References **13**

10 From Sensor to Digital Value

10.1 The Measurement Chain

This chapter puts all building blocks of the lecture together. We design a simple digital thermometer: a resistive temperature sensor measures $0\,^{\circ}\mathrm{C} \ldots 100\,^{\circ}\mathrm{C}$, and the result is shown as a 3-bit code on LEDs. Every stage of the chain is a circuit from an earlier chapter: the bridge from chapter 5, the instrumentation amplifier from chapter 6, the low-pass from chapter 7, the sample and hold from chapter 8 and the flash converter from chapter 9. The LED with its series resistor even goes back to chapter 1.
The design is done twice: on paper, with all calculations, and in Falstad, where the complete chain runs and can be checked against the paper values. The chapter is also the worked solution of last year's capstone task (exercise 10.1).

Five stages from sensor to code



- requirement: $0\,°\text{C} \ldots 100\,°\text{C} \rightarrow 0\,\text{V} \ldots 4.00\,\text{V} \rightarrow$ code 000 …111
- $4\,\text{V}$ full scale makes the step size round: $\text{LSB} = 4\,\text{V}/2^3 = 0.5\,\text{V} \mathrel{\widehat{=}} 12.5\,°\text{C}$
- interference: 50 Hz, 5 mV in series with the sensor (overhead lamps)

Specification and design values

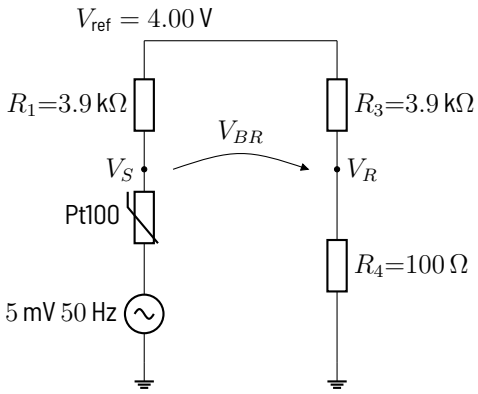
Quantity	Requirement	Design value
sensor	resistive, slider in Falstad	Pt100, $100\,\Omega \ldots 138.5\,\Omega$
sensor current	≤ 1 mA	1.000 mA … 0.990 mA
full scale	$0\,\text{V} \ldots 4.00\,\text{V}$	$G = 10 \cdot 10.76 = 107.6$
gain per stage	< 100	1, 10, 10.76
anti-aliasing	$f_c \approx 2.5$ Hz	$6.8\,\text{k}\Omega$, $10\,\mu\text{F}$: 2.34 Hz
sampling	$f_s = 5$ Hz	window 20 ms, $C_H = 1\,\mu\text{F}$
ADC	3 bit, LSB 0.5 V	$8 \times 1\,\text{k}\Omega$, 7 comparators
display	LED + resistor	$10\times$ (LED + $330\,\Omega$), 9.8 mA

▶ Falstad: `examples/capstone/capstone.txt`

10.2 Stage 1: Signal Conditioning

The sensor is a Pt100 platinum resistor with $R(T) = R_0\,(1 + \alpha T)$, $R_0 = 100\,\Omega$, $\alpha = 3.85 \times 10^{-3}/\text{K}$, so it changes from $100\,\Omega$ at $0\,°\text{C}$ to $138.5\,\Omega$ at $100\,°\text{C}$. We feed the bridge from the same $4.00\,\text{V}$ reference that later supplies the ADC ladder. If the reference is $1\,\%$ too high, the bridge output and all thresholds of the converter are $1\,\%$ too high as well, and the code does not change: the measurement is *ratiometric*. The reference only needs to be stable while it is shared, not accurate.

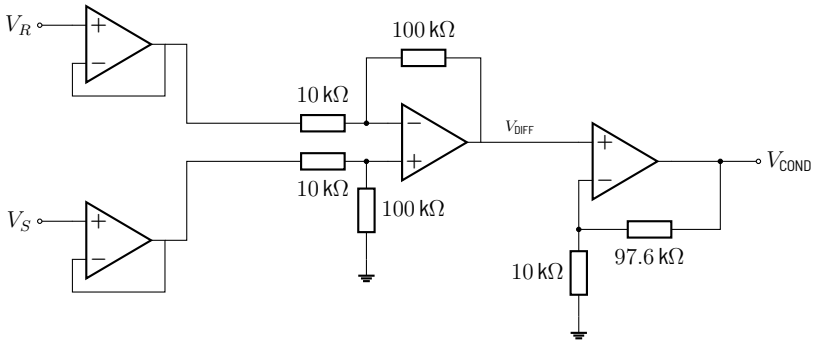
Bridge design



- sensor current $I_S = V_{\text{ref}}/(R_1 + R) = 4\,\text{V}/4000\,\Omega = 1.000\,\text{mA}$ at $0\,°\text{C}$ ($0.990\,\text{mA}$ at $100\,°\text{C}$)
- balanced at $0\,°\text{C}$: $R_4 = R(0\,°\text{C})$, so $V_{BR} = 0$
- $V_{BR}(T) = V_{\text{ref}}\left(\frac{R(T)}{R_1+R(T)} - \frac{R_4}{R_3+R_4}\right)$, $\quad V_{BR}(100\,°\text{C}) = 37.18\,\text{mV}$

The bridge output is not exactly linear in T, because the sensor current falls slightly with rising R. With $R_1 = 39\,R_0$ the error is small: at $50\,°\text{C}$ the output is $0.09\,\text{mV}$ higher than the straight line, $0.25\,\%$ of full scale, far below one LSB ($12.5\,\%$). A larger R_1 would reduce it further, but also the signal.

Instrumentation amplifier and gain stage



- buffers: input current zero, the bridge is not loaded
- difference amplifier: $V_{\text{DIFF}} = \frac{100\,\text{k}\Omega}{10\,\text{k}\Omega}(V_S - V_R) = 10\,V_{BR}$
- gain stage: $G_2 = 1 + \frac{97.6\,\text{k}\Omega}{10\,\text{k}\Omega} = 10.76$

Gain: how much, and in how many stages?

- required total gain $G = 4.00\,\text{V}/37.18\,\text{mV} = 107.6$
- split: $G_1 = 10$ (difference amplifier), $G_2 = 4.00/(10 \cdot 0.03718) = 10.758$
  - E96: $R_F = 97.6\,\text{k}\Omega \Rightarrow G_2 = 10.76$, full scale $4.0005\,\text{V}$
  - in hardware: $R_F = 91\,\text{k}\Omega + 10\,\text{k}\Omega$ trimmer, adjusted at $100\,°\text{C}$
- finite open-loop gain $A = 10^5$: relative gain error $\approx G/A$
  - one stage $G = 107.6$: $0.11\,\%$; two stages: $2 \times 0.01\,\%$
  - a single stage with $G > 1000$ would lose more than $1\,\%$: keep $G < 100$ per stage

Conditioning

$$V_{\mathrm{COND}} = G_1 G_2 \, V_{\mathrm{ref}} \left(\frac{R(T)}{R_1 + R(T)} - \frac{R_4}{R_3 + R_4} \right), \quad G_1 G_2 = 107.6$$

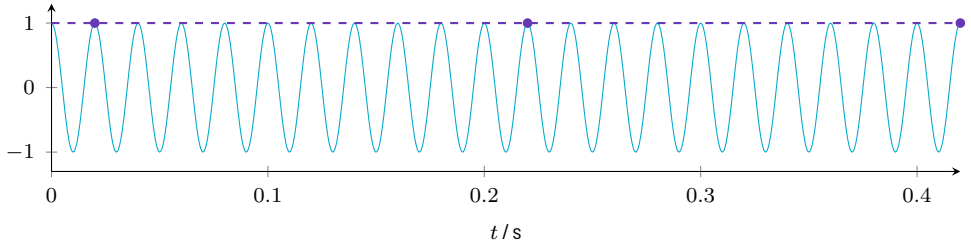
Splitting the gain over two stages has a second advantage: the zero point is set by the bridge (R_4) and the span by one resistor (R_F). The two adjustments do not influence each other, so the trimming procedure is simple: set the sensor to $100\,\Omega$ and check 0 V, then set it to $138.5\,\Omega$ and adjust R_F for 4.00 V.

10.3 Stage 2: Anti-Aliasing

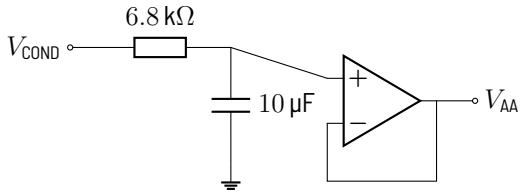
The temperature changes slowly, so five samples per second are plenty. The sampling theorem then limits the usable bandwidth to $f_s/2 = 2.5$ Hz. Everything above this frequency must be removed *before* the sampler, because after sampling it can no longer be distinguished from a real signal. The 50 Hz interference is the dangerous part: after the gain of 107.6 it has an amplitude of about half a volt, one full LSB.

Where does the 50 Hz go?

- sampling with f_s folds a frequency f to $f_{\text{alias}} = |f - k f_s|$ (nearest k)
- $f = 50$ Hz, $f_s = 5$ Hz: $|50 - 10 \cdot 5| = 0$ Hz
  - every sample sees the interference at the same phase
  - the result is a **constant offset**: nothing on the scope looks wrong
- amplitude at V_{COND}: $5\,\text{mV} \cdot \frac{R_1}{R_1+R} \cdot 107.6 = 0.52\,\text{V}$ (about 1 LSB)



Anti-aliasing low-pass



- $f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \cdot 6.8\,\text{k}\Omega \cdot 10\,\mu\text{F}} = 2.34\,\text{Hz} \le f_s/2$
- at 50 Hz: $|H| = 1/\sqrt{1 + (50/2.34)^2} = 0.047$ (-26.6 dB)
- residual offset: $0.52\,\text{V} \cdot 0.047 = 25\,\text{mV} < \text{LSB}/2 = 250\,\text{mV}$
- the follower decouples the RC network from the S&H input

For 3 bits a first-order filter is sufficient. For a 10-bit converter (LSB$/2 = 2$ mV) the residual 25 mV would be twelve times too large; a second- or third-order filter (chapter 7), a higher sampling rate followed by digital averaging, or a sampling rate that is *not* a sub-multiple of 50 Hz would then be needed.

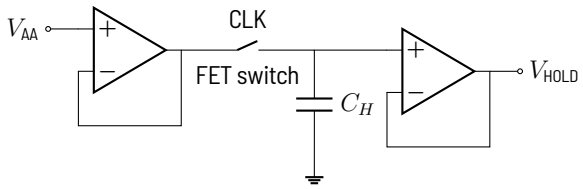
Design card: Nyquist & anti-aliasing (capstone values)

$f_s = 5$ Hz $\Rightarrow$ bandwidth ≤ 2.5 Hz; $f_{\text{alias}} = |f - k f_s|$; $f_c = 1/(2\pi RC) \le f_s/2$; residual $= \hat{v}\,/\sqrt{1 + (f/f_c)^2}$ must stay $<$ LSB$/2$.

10.4 Stage 3: Sample and Hold

The flash converter is fast and would not strictly need a sample and hold. We use one anyway, because it defines the sampling instant: the input of the converter changes only once per sampling period, so the LEDs show a stable code, and the aliasing experiment becomes visible. The circuit is the buffered S&H of chapter 8.

Buffered sample and hold



- clock 5 Hz, duty $10\,\%$: switch closed for 20 ms (track), open for 180 ms (hold)
- input buffer: charges C_H quickly; output buffer: no load on C_H

Choosing the hold capacitor

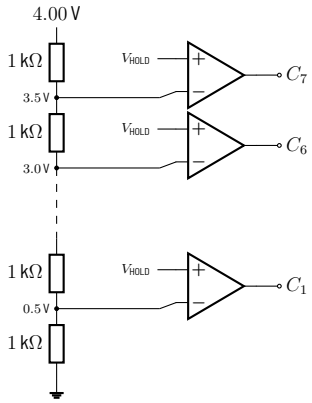
- **charging:** $\tau = R_{\text{on}} C_H = 20\,\Omega \cdot 1\,\mu\text{F} = 20\,\mu\text{s}$
  - settle to LSB$/2$ of 3 bits: $t = \tau \ln 2^4 = 2.8\,\tau = 55\,\mu\text{s} \ll 20$ ms
- **droop:** leakage $I_L = 1$ nA (switch + buffer input)
  - $\Delta V = I_L\, t_{\text{hold}} / C_H = 1\,\text{nA} \cdot 0.18\,\text{s} / 1\,\mu\text{F} = 0.18\,\text{mV}$
- both limits leave a wide range: 10 nF ...100 µF would work
  - choose a film capacitor with low leakage and dielectric absorption: $C_H = 1$ µF

In hardware a small resistor (e.g. 1 kΩ, as in the examples of chapter 8) is usually put in series with the switch: it limits the charging current that the input buffer must deliver and isolates the buffer from the capacitive load. Then $\tau = 1.02$ ms and settling to LSB$/2$ takes 2.8 ms, still well inside the window. The limits are far apart because the signal is slow and the resolution low. They move together quickly in a fast, high-resolution system: a 12-bit converter sampling at 1 MHz has a track time of a few hundred nanoseconds and needs settling to 2^{-13}, which is why real S&H stages use picofarad capacitors and low-resistance switches.

10.5 Stages 4 and 5: Flash ADC, Encoder and Display

The held voltage is compared with seven thresholds at once. The thresholds come from a ladder of eight equal resistors across the 4.00 V reference. Each comparator answers one question, "is the input above $k \cdot 0.5$ V?", and together they form a thermometer code: all comparators below the input voltage output 1, all above output 0. An encoder turns the seven bits into the 3-bit binary number.

Reference ladder and comparators



- ladder current $4\,\text{V}/8\,\text{k}\Omega = 0.5\,\text{mA}$; comparator inputs draw no current
- $C_k = 1$ if $V_{\text{HOLD}} > k \cdot 0.5\,\text{V}$; the number of ones is the level 0…7

Quantization and encoder

V_{HOLD} / V	level	C_7	C_6	C_5	C_4	C_3	C_2	C_1	B_2	B_1	B_0
0.0–0.5	0	0	0	0	0	0	0	0	0	0	0
0.5–1.0	1	0	0	0	0	0	0	1	0	0	1
1.0–1.5	2	0	0	0	0	0	1	1	0	1	0
1.5–2.0	3	0	0	0	0	1	1	1	0	1	1
2.0–2.5	4	0	0	0	1	1	1	1	1	0	0
2.5–3.0	5	0	0	1	1	1	1	1	1	0	1
3.0–3.5	6	0	1	1	1	1	1	1	1	1	0
3.5–4.0	7	1	1	1	1	1	1	1	1	1	1

- $B_2 = C_4$ (level ≥ 4)
- $B_1 = C_2\overline{C_4} + C_6$ (levels 2, 3, 6, 7)
- $B_0 = C_1\overline{C_2} + C_3\overline{C_4} + C_5\overline{C_6} + C_7$ (odd levels: the edge of the thermometer is at an odd position)

Display

- 7 red LEDs from $C_1 \ldots C_7$: bar graph, a thermometer you can read without decoding
- 3 green LEDs from $B_2B_1B_0$: the binary result
- each LED with its own series resistor (chapter 1), driven by a 5 V logic output: $R = \frac{5\,\text{V} - V_F}{I_F} = \frac{5\,\text{V} - 1.8\,\text{V}}{10\,\text{mA}} = 320\,\Omega \rightarrow$ E12: $330\,\Omega$
  - Falstad: $I_F = 9.8\,\text{mA}$, $V_F = 1.78\,\text{V}$
- never one common resistor for several LEDs: the current would depend on how many are on

10.6 The Complete Chain in Falstad

The Falstad model contains all five stages in labelled boxes. The stages are connected with labeled nodes (`VCOND`, `VAA`, `VHOLD`, `C1`…`C7`, `B0`…`B2`); nodes with the same name are connected, which keeps the drawing readable. The sensor resistance is set with the slider "Sensor R". The first scope shows the filtered signal and the held staircase, the second the conditioned signal and the held value.

Using the model

- open ▶ Falstad: `examples/capstone/capstone.txt`
- move the slider from $100\,\Omega$ to $138.5\,\Omega$: the bar graph fills up, the binary LEDs count
- hover over a label to read its voltage; the held value changes every 200 ms
- after a slider step, wait about 0.5 s: the low-pass needs $5\tau = 5 \cdot 68$ ms

Verification table

For each test point the sensor resistance is computed from the conditioning equation so that V_{COND} lies in the centre of a code bin. The "Falstad" columns are the values of the simulation (with the 50 Hz interference switched on) just before the next sample.

T / °C	R / Ω	V_{COND} target	V_{HOLD} Falstad	code expected	code Falstad
6.2	102.38	0.25 V	0.251 V	000	000
18.6	107.16	0.75 V	0.751 V	001	001
43.5	116.75	1.75 V	1.749 V	011	011
56.0	121.56	2.25 V	2.250 V	100	100
81.1	131.22	3.25 V	3.249 V	110	110
93.7	136.07	3.75 V	3.750 V	111	111

- the temperatures differ slightly from the linear mapping $25\,°\mathrm{C/V}$ (6, 19, 44, 56, 81, $94\,°$C) because of the bridge nonlinearity

The aliasing experiment

- ▶ Falstad: `examples/capstone/capstone_nofilter.txt`: the S&H takes V_{COND} directly
- V_{COND} carries 0.52 V of 50 Hz; the S&H always samples at the same phase

	V_{COND} (DC)	V_{HOLD}	code
without filter	1.75 V	2.27 V	100 (wrong)
without filter	0.25 V	0.77 V	001 (wrong)
with filter	1.75 V	1.75 V	011

- without the filter the held value is a clean staircase, just 0.52 V too high: one full code too high
- the size of the offset depends on the phase between mains and sampling clock (here the worst case)

10.7 Exercises

Exercise 10.1: From sensor to digital value (capstone task 2025/26)

Build a complete data-acquisition chain in Falstad that turns a slowly changing sensor signal ($0\,°$C … $100\,°$C) into a 3-bit code. Label the five stages Conditioning | Anti-aliasing | Sample & Hold | Flash ADC | Encoder & Display.

a) Conditioning: resistive sensor in a Wheatstone bridge, sensor current ≤ 1 mA, sensor as a resistor with a slider; instrumentation-amplifier front end (two unity-gain buffers and a difference amplifier); gain trimmed to 0 V at the bottom and exactly 4.00 V at the top of the range; less than 100 per stage.

b) Anti-aliasing: sampling clock $f_s = 5$ Hz. Add the 50 Hz, 5 mV noise source in series with the sensor. Observe the held value without a filter, then add an active low-pass with $f_c \approx 2.5$ Hz and justify f_c from the Nyquist criterion.

c) Sample and hold at 5 Hz: follower, FET switch, hold capacitor, follower. Choose the hold capacitor.

d) Flash ADC: ladder $8 \times 1\,\text{k}\Omega$ from 4 V, seven comparators, thermometer code.

e) Encoder ($B_2 = C_4$, $B_1 = C_2\overline{C_4} + C_6$, $B_0 = C_1\overline{C_2} + C_3\overline{C_4} + C_5\overline{C_6} + C_7$) and display: 7-LED bar graph and three binary LEDs, each LED with its own resistor.

f) Verification table for six inputs near the centres of different code bins, and one line on the aliasing experiment.

Solution 10.1

The design of this chapter is one possible solution; the complete model is
▶ Falstad: `examples/capstone/capstone.txt`.

a) Pt100 with $R_1 = R_3 = 3.9\,\text{k}\Omega$, $R_4 = 100\,\Omega$ at $V_{\text{ref}} = 4.00$ V: $I_S = 1.000$ mA, bridge output 0 mV … 37.18 mV. Buffers, difference amplifier $G_1 = 10$, non-inverting stage $G_2 = 1 + 97.6\,\text{k}\Omega/10\,\text{k}\Omega = 10.76$: $V_{\text{COND}} = 0$ V … 4.00 V.

b) Without filter the 50 Hz aliases to $|50 - 10 \cdot 5| = 0$ Hz; the held value is offset by up to 0.52 V (Falstad: 1.75 V becomes 2.27 V, code 100 instead of 011, ▶ Falstad: `examples/capstone/capstone_nofilter.txt`). Filter $R = 6.8\,\text{k}\Omega$, $C = 10\,\mu$F, $f_c = 2.34$ Hz $\leq f_s/2$, 50 Hz attenuated to $4.7\,\%$: 25 mV residual, below LSB$/2$.

c) $C_H = 1\,\mu$F: $\tau = 20\,\mu$s charges within the 20 ms window; droop 0.18 mV per hold period for 1 nA leakage.

d) Taps at $0.5, 1.0, \ldots, 3.5$ V; ladder current 0.5 mA.

e) 3 inverters, 4 AND gates, 2 OR gates (one with 4 inputs); 10 LEDs with $330\,\Omega$, 9.8 mA each.

f) See the verification table above. Aliasing line: "Without the anti-aliasing filter the held value showed a constant offset of 0.52 V (one code too high); with the 2.34 Hz low-pass the offset fell below 30 mV and all codes were correct."

Exercise 10.2: Ratiometric measurement

The 4.00 V reference of the capstone is in fact 4.10 V.

a) How large is V_{COND} at $100\,°$C?

b) Which code does the converter show at $100\,°$C and at $56\,°$C? Compare with a design in which the bridge is supplied from a separate, exact 4.00 V source.

Solution 10.2

a) The bridge output scales with the reference: $V_{\text{COND}} = 4.0005\,\text{V} \cdot 4.10/4.00 = 4.10$ V.

b) The thresholds scale as well ($k \cdot 0.5125$ V). The ratio $V_{\text{COND}}/V_{\text{ref}}$ is unchanged, so both codes are unchanged: 111 at $100\,°$C, 100 at $56\,°$C ($V_{\text{COND}} = 2.25\,\text{V} \cdot 1.025 = 2.306$ V, threshold of level 4: 2.05 V, of level 5: 2.5625 V). With an exact bridge supply and a 4.10 V ladder the thresholds would be $2.5\,\%$ too high while the signal is correct; a value just above a threshold (e.g. 2.02 V) would then be read one code too low. Ratiometric operation removes the reference from the error budget.

Exercise 10.3: Another sampling rate

The sampling clock of the capstone is changed to $f_s = 4$ Hz and the anti-aliasing filter is removed.

a) To which frequency does the 50 Hz interference alias?

b) Describe what the LEDs show for a constant temperature with $V_{\text{COND}} = 1.75$ V.

c) Is this better or worse than the 0 Hz alias at 5 Hz?

Solution 10.3

a) $|50 - 12 \cdot 4| = 2$ Hz, exactly the Nyquist frequency $f_s/2$: consecutive samples see the interference with a phase step of $2\pi \cdot 50/4 = 25\pi$, so its sign alternates from sample to sample.

b) The held value alternates between $1.75 + \hat{v}\sin\varphi$ and $1.75 - \hat{v}\sin\varphi$ ($\hat{v} = 0.52$ V): the LEDs blink at 2 Hz between two codes, e.g. 100 and 010 in the worst case, or stay at 011 if the phase happens to be $\varphi \approx 0$.

c) It is visible, so it is "better" than a silent offset, but the reading is still wrong. Only the low-pass in front of the sampler removes the error for every sampling rate.

Exercise 10.4: Four bits

The capstone shall be extended to 4 bits with the same 4.00 V full scale.

a) What is the new LSB, in volts and in °C? How many comparators and ladder resistors are needed?

b) Is the first-order anti-aliasing filter still sufficient?

c) Is the bridge nonlinearity still acceptable?

Solution 10.4

a) LSB $= 4\,\text{V}/16 = 0.25\,\text{V} \mathrel{\widehat{=}} 6.25\,°\text{C}$; $2^4 - 1 = 15$ comparators, 16 resistors (taps at $k \cdot 0.25$ V).

b) Residual 25 mV $<$ LSB$/2 = 125$ mV: yes, still a factor five of margin.

c) The maximum deviation from the straight line is about $0.09\,\text{mV} \cdot 107.6 = 10$ mV $\ll 125$ mV: yes.

Exercise 10.5: Hold capacitor at the limits

In the capstone S&H the analog switch has $R_{\text{on}} = 1\,\text{k}\Omega$ and the total leakage current is 10 nA. Determine the smallest and the largest usable hold capacitor if the droop during one hold period and the remaining charging error at the end of the window shall both stay below LSB$/4 = 125$ mV for a full-scale step.

Solution 10.5

Droop: $I_L\, t_{\text{hold}}/C_H < 0.125\,\text{V} \Rightarrow C_H > 10\,\text{nA} \cdot 0.18\,\text{s}/0.125\,\text{V} = 14.4$ nF. Charging: $4\,\text{V} \cdot e^{-t_w/\tau} < 0.125\,\text{V}$ $\Rightarrow t_w > \tau \ln 32 = 3.47\,\tau$, so $\tau < 20\,\text{ms}/3.47 = 5.8$ ms and $C_H < 5.8\,\text{ms}/1\,\text{k}\Omega = 5.8\,\mu\text{F}$. Any value between about 15 nF and 5.8 µF works; 1 µF is a good choice in the middle (on a logarithmic scale). In Falstad set the switch's on resistance to $1\,\text{k}\Omega$ and C_H to 10 µF in

▶ Falstad: `examples/capstone/capstone.txt`: after a slider step the held staircase needs several samples to reach the new value.

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